

Linear Algebra Done Right

Reading Notes

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Contents

1	Vector Spaces	2
1.1	$\mathbb{R}^n$ and $\mathbb{C}^n$	2
1.1.1	Complex Numbers, $\mathbb{C}$	2
1.1.2	Lists	2
1.1.3	$\mathbb{F}^n$	2
1.2	Definition of a Vector Space	3
1.2.1	Some Property Proofs	3
1.3	Subspaces	4
1.3.1	Sums of Subspaces	4
1.3.2	Direct Sums	5
1.4	Exercise 1A	5
1.5	Exercise 1B	8
1.6	Exercise 1C	10

1 Vector Spaces

1.1 $\mathbb{R}^n$ and $\mathbb{C}^n$

Before defining vector spaces, Axler establishes the fields we are working over. The symbol $\mathbb{F}$ will always denote either the real numbers $\mathbb{R}$ or the complex numbers $\mathbb{C}$.

Definition 1.1 (Fields). *A field is a set containing at least two distinct elements called 0 and 1, along with operations of addition and multiplication satisfying the properties of commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity.*

1.1.1 Complex Numbers, $\mathbb{C}$

Definition 1.2 (Complex Numbers).

- A complex number is an ordered pair (a, b) where $a, b \in \mathbb{R}$, but we will write this as $a + bi$.
- The set of all complex numbers is denoted by $\mathbb{C}$:

$$\mathbb{C} = \{a + bi : a, b \in \mathbb{R}\}.$$

- Addition and multiplication on $\mathbb{C}$ are defined by

$$(a + bi) + (c + di) = (a + c) + (b + d)i,$$

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i;$$

here $a, b, c, d \in \mathbb{R}$

My thoughts

Just assume $i^2 = -1$ and everything works the same as in $\mathbb{R}$. Commutativity, associativity, identities, additive inverse, multiplicative inverse and distributivity are all properties of complex arithmetic.

1.1.2 Lists

Definition 1.3 (List, length).

- Suppose n is a nonnegative integer. A list of length n is an ordered collection of n elements (which might be numbers, other lists, or more abstract objects).
- Two lists are equal if and only if they have the same length and the same elements in the same order.

My thoughts

A list can have multiple same entries unlike sets. Fields and Vector Spaces (defined later) are sets, but a list is a tuple.

1.1.3 $\mathbb{F}^n$

Definition 1.4 ($\mathbb{F}^n$, coordinate). $\mathbb{F}^n$ is the set of all lists of length n of elements of $\mathbb{F}$:

$$\mathbb{F}^n = \{(x_1, \dots, x_n) : x_k \in \mathbb{F} \text{ for } k \in \{1, \dots, n\}\}$$

For $(x_1, \dots, x_n) \in \mathbb{F}^n$ and $k \in \{1, \dots, n\}$, we say that x_k is the k^{th} coordinate of $(x_1, \dots, x_n)$.

Definition 1.5 (addition in F^n). *Addition in F^n is defined by adding corresponding coordinates:*

$$(x_1, \dots, x_n) + (y_1, \dots, y_n) = (x_1 + y_1, \dots, x_n + y_n)$$

Definition 1.6 (additive inverse in $\mathbb{F}^n$, $-x$). *For $x \in \mathbb{F}^n$, the additive inverse of x , denoted by $-x$, is the vector $-x \in \mathbb{F}^n$ such that*

$$x + (-x) = 0$$

Thus if $x = (x_1, \dots, x_n)$, then $-x = (-x_1, \dots, -x_n)$.

Definition 1.7 (scalar multiplication in $\mathbb{F}^n$). *The product of a number λ and a vector in $\mathbb{F}^n$ is computed by multiplying each coordinate of the vector by λ :*

$$\lambda(x_1, \dots, x_n) = (\lambda x_1, \dots, \lambda x_n);$$

here $\lambda \in \mathbb{F}$ and $(x_1, \dots, x_n) \in \mathbb{F}^n$

1.2 Definition of a Vector Space

Now we abstract the properties of $\mathbb{R}^n$ and $\mathbb{C}^n$ into a general structure.

Definition 1.8 (Vector Space). *A vector space is a set $\mathcal{V}$ along with an addition on $\mathcal{V}$ and a scalar multiplication on $\mathcal{V}$ such that commutativity, associativity, identities, additive inverse, and distributivity hold.*

My thoughts

Essentially, the difference between a field and a vector space is the fact that only scalar multiplication is defined for elements of a vector space. This also implies it is not necessary for a vector space to have multiplicative inverses. Also note, a field is a standalone set while a vector space relies on an underlying field to exist at all.

Definition 1.9 (Real Vector Space, Complex Vector Space).

- *A vector space over $\mathbb{R}$ is called a real vector space.*
- *A vector space over $\mathbb{C}$ is called a complex vector space.*

1.2.1 Some Property Proofs

Theorem 1.10 (Unique Additive Identity). *A vector space has a unique additive identity.*

Proof. Suppose 0 and $0'$ are both additive identities for V . Then

$$0' = 0' + 0 = 0 + 0' = 0,$$

where the first equality holds because 0 is an additive identity, the second equality comes from commutativity, and the third equality holds because $0'$ is an additive identity. Thus $0' = 0$, proving that $\mathcal{V}$ has only one additive identity. $\square$

Theorem 1.11. *a number times the vector 0 is 0*

Proof. For $v \in \mathcal{V}$, we have $0v = (0 + 0)v = 0v + 0v$. Adding the additive inverse of $0v$ to both sides of the equation above gives $0 = 0v$, as desired. $\square$

Theorem 1.12. *the number -1 times a vector gives the additive inverse of the vector*

Proof. For $v \in \mathcal{V}$, we have $v + (-1)v = 1v + (-1)v = (1 + (-1))v = 0v = 0$. This equation says that $(-1)v$, when added to v , gives 0 . Thus $(-1)v$ is the additive inverse of v , as desired $\square$

1.3 Subspaces

Definition 1.13 (Subspace). A subset U of $\mathcal{V}$ is called a subspace of $\mathcal{V}$ if U is also a vector space under the same addition and scalar multiplication.

1.3.1 Sums of Subspaces

Definition 1.14 (Sum of Subspaces). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. The sum of $\mathcal{V}_1, \dots, \mathcal{V}_m$, denoted by $\mathcal{V}_1 + \dots + \mathcal{V}_m$, is the set of all possible sums of elements of $\mathcal{V}_1, \dots, \mathcal{V}_m$. More precisely,

$$\mathcal{V}_1 + \dots + \mathcal{V}_m = \{v_1 + \dots + v_m : v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m\}$$

Theorem 1.15. Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

My thoughts

I will add my own proof that sum of subspaces is a subspace (although Axler skips this) because I think it is necessary.

Proof. Let us first prove that sum of subspaces is a subspace

- $\mathcal{V}_1 + \dots + \mathcal{V}_m$ contains 0 (additive identity) since all the subspaces contain 0 and sum of subspaces contain every possible sum of their elements.
- Let $x, y \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$.
By definition we can express them as $x = v_1 + \dots + v_m$ and $y = u_1 + \dots + u_m$ such that $u_1, v_1 \in \mathcal{V}_1, \dots, u_m, v_m \in \mathcal{V}_m$. So we have

$$x + y = (v_1 + \dots + v_m) + (u_1 + \dots + u_m)$$

By commutativity and associativity of vector addition,

$$x + y = (v_1 + u_1) + \dots + (v_m + u_m)$$

Since $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces, $(v_1 + u_1) \in \mathcal{V}_1, \dots, (v_m + u_m) \in \mathcal{V}_m$

Hence we conclude $(x + y) \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, it is closed under addition

- Let $x \in (\mathcal{V}_1 + \dots + \mathcal{V}_m)$. and $x = v_1 + \dots + v_m$ where $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. Let c be any scalar from the field $\mathbb{F}$.
Multiplying x with the scalar c yields:

$$cx = c(v_1 + \dots + v_m) = cv_1 + \dots + cv_m$$

Since $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces $cv_1 \in \mathcal{V}_1, \dots, cv_m \in \mathcal{V}_m$.

We conclude $cx \in \mathcal{V}_1 + \dots + \mathcal{V}_m$

Thus, sum of subspaces is a subspace.

Now we can start with the proof for $\mathcal{V}_1, \dots, \mathcal{V}_m$ being the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$.

The subspaces $\mathcal{V}_1, \dots, \mathcal{V}_m$ are all contained in $\mathcal{V}_1 + \dots + \mathcal{V}_m$, considering sums $v_1 + \dots + v_m$ where all except one of the v_k 's are 0.

Conversely, every subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$ contains $\mathcal{V}_1 + \dots + \mathcal{V}_m$ (because subspaces must contain all finite sums of their elements). Thus $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is the smallest subspace of $\mathcal{V}$ containing $\mathcal{V}_1, \dots, \mathcal{V}_m$. $\square$

1.3.2 Direct Sums

Definition 1.16 (Direct Sum). Suppose $\mathcal{V}_1, \dots, \mathcal{V}_m$ are subspaces of $\mathcal{V}$.

- The sum $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is called a *direct sum* if each element of $\mathcal{V}_1 + \dots + \mathcal{V}_m$ can be written in only one way as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$.
- If $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, then $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_m$ denotes $\mathcal{V}_1 + \dots + \mathcal{V}_m$, with the $\oplus$ notation serving as an indication that this is a direct sum.

Theorem 1.17. Suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ are subspaces of $\mathcal{V}$. Then $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum if and only if the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

Proof. First suppose $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum. Then the definition of direct sum implies that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0.

My thoughts

This is because taking every $v_k = 0$ is a trivial solution, but since we know it is a direct sum, the only solution would be taking every $v_k = 0$.

Now suppose that the only way to write 0 as a sum $v_1 + \dots + v_m$, where each $v_k \in \mathcal{V}_k$, is by taking each v_k equal to 0. To show that $\mathcal{V}_1 + \dots + \mathcal{V}_m$ is a direct sum, let $v \in \mathcal{V}_1 + \dots + \mathcal{V}_m$. We can write

$$v = v_1 + \dots + v_m$$

for some $v_1 \in \mathcal{V}_1, \dots, v_m \in \mathcal{V}_m$. To show that this representation is unique, suppose we also have

$$v = u_1 + \dots + u_m$$

where $u_1 \in \mathcal{V}_1, \dots, u_m \in \mathcal{V}_m$. Subtracting these two equations, we have

$$0 = (v_1 - u_1) + \dots + (v_m - u_m)$$

Because $v_1 - u_1 \in \mathcal{V}_1, \dots, v_m - u_m \in \mathcal{V}_m$, the equation above implies that each $v_k - u_k$ equals 0. Thus $v_1 = u_1, \dots, v_m = u_m$, as desired. $\square$

Theorem 1.18. Suppose U and W are subspaces of V . Then

$$U + W \text{ is a direct sum} \iff U \cap W = \{0\}.$$

Proof. First suppose that $U + W$ is a direct sum. If $v \in U \cap W$, then $0 = v + (-v)$, where $v \in U$ and $-v \in W$. By the unique representation of 0 as the sum of a vector in U and a vector in W , we have $v = 0$. Thus $U \cap W = \{0\}$, completing the proof in one direction. To prove the other direction, now suppose $U \cap W = \{0\}$. To prove that $U + W$ is a direct sum, suppose $u \in U, w \in W$, and $0 = u + w$. To complete the proof, we only need to show that $u = w = 0$. The equation above implies that $u = -w \in W$. Thus $u \in U \cap W$. Hence $u = 0$, which by the equation above implies that $w = 0$, completing the proof. $\square$

1.4 Exercise 1A

My thoughts

I can't find a way to prove these identities by themselves, maybe I need to learn analysis for that. I will just be assuming $\mathbb{R}$ is already known to be a field, and use those properties to prove these.

Question 1. Show that $\alpha + \beta = \beta + \alpha$ for all $\alpha, \beta \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, $\alpha + \beta = a + ib + c + id$

$$\begin{aligned} &= (a + c) + (b + d)i \\ &= (c + a) + (d + b)i \\ &= (c + id) + (a + ib) \\ &= \beta + \alpha \end{aligned}$$

□

Question 2. Show that $(\alpha + \beta) + \lambda = \alpha + (\beta + \lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, and $\lambda = e + fi$ (where $a, b, c, d, e, f \in \mathbb{R}$), the left side of the equation expands as:

$$\begin{aligned} &(\alpha + \beta) + \lambda \\ &= ((a + c) + (b + d)i) + (e + fi) \\ &= ((a + c) + e) + ((b + d) + f)i \end{aligned}$$

Because addition is associative in the real numbers, we can rewrite the real and imaginary components as $(a + (c + e))$ and $(b + (d + f))$.

Expanding the right side gives:

$$\begin{aligned} &\alpha + (\beta + \lambda) \\ &= (a + bi) + ((c + e) + (d + f)i) \\ &= (a + (c + e)) + (b + (d + f))i \end{aligned}$$

□

Question 3. Show that $(\alpha\beta)\lambda = \alpha(\beta\lambda)$ for all $\alpha, \beta, \lambda \in \mathbb{C}$.

Proof. Letting $\alpha = a + bi$, $\beta = c + di$, and $\lambda = e + fi$ (where $a, b, c, d, e, f \in \mathbb{R}$), the left side of the equation expands as:

$$((ac - bd) + (ad + bc)i)(e + fi)$$

Multiplying this out yields a real part of $ace - bde - adf - bcf$ and an imaginary part of $acf - bdf + ade + bce$.

Expanding the right side $\alpha(\beta\lambda)$ gives

$$(a + bi)((ce - df) + (cf + de)i)$$

Multiplying this out yields a real part of $ace - adf - bcf - bde$ and an imaginary part of $acf + ade + bce - bdf$. □

Question 4. Show that $\lambda(\alpha + \beta) = \lambda\alpha + \lambda\beta$ for all $\lambda, \alpha, \beta \in \mathbb{C}$

Proof. Expanding $\lambda(\alpha + \beta)$ with $\lambda = e + fi$, $\alpha = a + bi$, and $\beta = c + di$ gives a real part of $ea + ec - fb - fd$ and an imaginary part of $eb + ed + fa + fc$. Multiplying out $\lambda\alpha + \lambda\beta$ yields the exact same terms, relying on the distributive property of real numbers to match them up. □

Question 5. Show that for every $\alpha \in \mathbb{C}$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha + \beta = 0$.

Proof. Setting $\alpha + \beta = 0$ with $\alpha = a + bi$ and $\beta = c + di$ results in $(a + c) + (b + d)i = 0$. Equating the real and imaginary components to zero means $a + c = 0$ and $b + d = 0$, yielding exactly $c = -a$ and $d = -b$. Because every real number has a uniquely defined additive inverse in $\mathbb{R}$, the components c and d exist and are unique, making β unique. □

Question 6. Show that for every $\alpha \in \mathbb{C}$ with $\alpha \neq 0$, there exists a unique $\beta \in \mathbb{C}$ such that $\alpha\beta = 1$.

Proof. $\alpha = a + bi$, $\beta = c + di$

Setting $(a + bi)(c + di) = 1$ results in $(ac - bd) + (ad + bc)i = 1 + 0i$. Equating the real and imaginary parts gives a system of two linear equations: $ac - bd = 1$ and $ad + bc = 0$. Since $\alpha \neq 0$, we know that a and b are not both zero, meaning $a^2 + b^2 \neq 0$. Solving this system for c and d yields exactly $c = \frac{a}{a^2+b^2}$ and $d = \frac{-b}{a^2+b^2}$. This produces a single, well-defined real value for both c and d . $\square$

Question 7. Show that $\frac{-1+\sqrt{3}i}{2}$ is a cube root of 1 (meaning that its cube equals 1).

Proof. Squaring $\frac{-1+\sqrt{3}i}{2}$ yields $\frac{-1-\sqrt{3}i}{2}$. Multiplying that squared result by the original term $\frac{-1+\sqrt{3}i}{2}$ gives $\frac{(-1)^2 - (\sqrt{3}i)^2}{4} = \frac{1 - (-3)}{4}$, which reduces perfectly to 1. $\square$

Question 8. Find two distinct square roots of i .

Proof. $\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}}i$ and $-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}}i$ $\square$

Question 9. Find $x \in \mathbb{R}^4$ such that $(4, -3, 1, 7) + 2x = (5, 9, -6, 8)$.

Proof. $(1/2, 6, -7/2, 1/2)$ $\square$

Question 10. Explain why there does not exist $\lambda \in \mathbb{C}$ such that $\lambda(2 - 3i, 5 + 4i, -6 + 7i) = (12 - 5i, 7 + 22i, -32 - 9i)$.

Proof. Setting $\lambda(2 - 3i) = 12 - 5i$ for the first component gives $\lambda = \frac{12-5i}{2-3i}$.

Multiplying the numerator and denominator by the conjugate $(2 + 3i)$ yields $\frac{24+36i-10i+15}{13}$, which simplifies to $\lambda = 3 + 2i$.

Testing this λ on the second component yields $(3 + 2i)(5 + 4i) = 15 + 12i + 10i - 8 = 7 + 22i$, which matches the required vector perfectly.

Now, applying $\lambda = 3 + 2i$ to the third component gives $(3 + 2i)(-6 + 7i) = -18 + 21i - 12i - 14 = -32 + 9i$. But the required third component in the target vector is $-32 - 9i$. $\square$

Question 11. Show that $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}^n$.

Proof. Let $x = (x_1, \dots, x_n)$, $y = (y_1, \dots, y_n)$, and $z = (z_1, \dots, z_n)$. Adding component-wise, the left side gives $((x_1 + y_1) + z_1, \dots, (x_n + y_n) + z_n)$. Because addition is associative in the underlying field $\mathbb{F}$, each component $(x_i + y_i) + z_i$ is equal to $x_i + (y_i + z_i)$. This perfectly constructs the right side $x + (y + z)$. $\square$

Question 12. Show that $(ab)x = a(bx)$ for all $x \in \mathbb{F}^n$ and all $a, b \in \mathbb{F}$.

Proof. Expanding x as $(x_1, \dots, x_n)$ means the left side $(ab)x$ becomes $((ab)x_1, \dots, (ab)x_n)$. The right side $a(bx)$ evaluates to $a(bx_1, \dots, bx_n)$, which then scales to $(a(bx_1), \dots, a(bx_n))$. Because multiplication within the underlying field $\mathbb{F}$ is associative, $(ab)x_i = a(bx_i)$ for every component $\square$

Question 13. Show that $1x = x$ for all $x \in \mathbb{F}^n$.

Proof. Expanding x into its components $(x_1, \dots, x_n)$ gives $1x = (1x_1, \dots, 1x_n)$. Because 1 is the multiplicative identity in the field $\mathbb{F}$, $1x_i = x_i$ for every individual component. This gives us the original vector x . $\square$

Question 14. Show that $\lambda(x + y) = \lambda x + \lambda y$ for all $\lambda \in \mathbb{F}$ and all $x, y \in \mathbb{F}^n$.

Proof. Expanding $\lambda(x + y)$ component-wise gives $(\lambda(x_1 + y_1), \dots, \lambda(x_n + y_n))$. Because the underlying field $\mathbb{F}$ has the distributive property, each component expands to $\lambda x_i + \lambda y_i$. This perfectly matches the component-wise expansion of the right side, $\lambda x + \lambda y = (\lambda x_1 + \lambda y_1, \dots, \lambda x_n + \lambda y_n)$. $\square$

Question 15. Show that $(a + b)x = ax + bx$ for all $a, b \in \mathbb{F}$ and all $x \in \mathbb{F}^n$.

Proof. Expanding $(a + b)x$ component-wise yields $((a + b)x_1, \dots, (a + b)x_n)$. Applying the distributive property of the underlying field $\mathbb{F}$ to each component gives $(ax_1 + bx_1, \dots, ax_n + bx_n)$. This perfectly matches the right side $ax + bx$ when you add the scaled vectors $(ax_1, \dots, ax_n)$ and $(bx_1, \dots, bx_n)$ together. $\square$

1.5 Exercise 1B

Question 1. Prove that $-(-v) = v$ for every $v \in \mathcal{V}$.

Proof.

$$\begin{aligned} -(-v) &= -(-v) + 0 \\ -(-v) &= -(-v) + (-v + v) \\ -(-v) &= (-(-v) + -v) + v \\ -(-v) &= 0 + v \\ -(-v) &= v \end{aligned}$$

$\square$

Question 2. Suppose $a \in \mathbb{F}, v \in \mathcal{V}$, and $av = 0$. Prove that $a = 0$ or $v = 0$.

Proof. Given equation:

$$av = 0$$

Since $a \in \mathbb{F}$ (a field) and $a \neq 0$, it has a multiplicative inverse $a^{-1} \in \mathbb{F}$.

$$\begin{aligned} a^{-1}(av) &= a^{-1}(0) \\ a^{-1}(av) &= 0 \end{aligned}$$

By associativity of scalar multiplication:

$$\begin{aligned} (a^{-1}a)v &= 0 \\ 1v &= 0 \\ v &= 0 \end{aligned}$$

If $a \neq 0$, then v must be 0. Therefore, if $av = 0$, it must be true that either $a = 0$ or $v = 0$. $\square$

Question 3. Suppose $v, w \in \mathcal{V}$. Explain why there exists a unique $x \in \mathcal{V}$ such that $v + 3x = w$.

Proof. First we prove existence:

$$x = \frac{1}{3}(w - v)$$

Since a vector space is closed under addition and scalar multiplication $x \in \mathcal{V}$

Now for uniqueness:

Suppose there are two vectors, x_1 and x_2 in V , that both satisfy the equation.

$$\begin{aligned} v + 3x_1 &= v + 3x_2 \\ -v + (v + 3x_1) &= -v + (v + 3x_2) \end{aligned}$$

Using associativity and additive inverse properties,

$$\begin{aligned} 3x_1 &= 3x_2 \\ x_1 &= x_2 \end{aligned}$$

Thus x is unique. $\square$

Question 4. The empty set is not a vector space. The empty set fails to satisfy only one of the requirements listed in the definition of a vector space. Which one?

Proof. It doesn't have the zero element. $\square$

Question 5. Show that in the definition of a vector space, the additive inverse condition can be replaced with the condition that $0v = 0$ for all $v \in \mathcal{V}$.

Proof. Let's assume the additive inverse condition does not exist. Instead we have $0v = 0$ for all $v \in \mathcal{V}$. We can arrive at the additive inverse criterion from here.

$$0v = 0$$

$$(1 + (-1))v = 0$$

Using distributivity,

$$1v + (-1)v = 0$$

Using multiplicative identity,

$$v + (-1)v = 0$$

The element $(-1)v$ satisfies the exact requirements of an additive inverse. $\square$

Question 6. Let ∞ and $-\infty$ denote two distinct objects, neither of which is in $\mathbb{R}$. Define an addition and scalar multiplication on $\mathbb{R} \cup \{\infty, -\infty\}$ as you could guess from the notation. Specifically, the sum and product of two real numbers is as usual, and for $t \in \mathbb{R}$ define

$$t\infty = \begin{cases} -\infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ \infty & \text{if } t > 0, \end{cases} \quad t(-\infty) = \begin{cases} \infty & \text{if } t < 0, \\ 0 & \text{if } t = 0, \\ -\infty & \text{if } t > 0, \end{cases}$$

and

$$\begin{aligned} t + \infty &= \infty + t = \infty + \infty = \infty, \\ t + (-\infty) &= (-\infty) + t = (-\infty) + (-\infty) = -\infty, \\ \infty + (-\infty) &= (-\infty) + \infty = 0. \end{aligned}$$

With these operations of addition and scalar multiplication, is $\mathbb{R} \cup \{\infty, -\infty\}$ a vector space over $\mathbb{R}$? Explain.

Proof. It is not a vector space.

If $t + \infty = \infty$, we have:

$$\begin{aligned} 1 + \infty &= 2 + \infty \\ 1 &= 2 \end{aligned}$$

We can cancel out the ∞ on both sides due to existence of additive inverse. $\square$

Question 7. Suppose S is a nonempty set. Let $\mathcal{V}^S$ denote the set of functions from S to $\mathcal{V}$. Define a natural addition and scalar multiplication on $\mathcal{V}^S$, and show that $\mathcal{V}^S$ is a vector space with these definitions.

Proof. Because $\mathcal{V}$ is a vector space, $\mathcal{V}^S$ will obey all vector space axioms through point-wise evaluations for every $x \in S$.

- Commutativity & Associativity: $(f + g)(x) = f(x) + g(x) = g(x) + f(x) = (g + f)(x)$. Associativity follows the exact same logic.

- Additive Identity: The zero function, defined by $0(x) = 0$ for all $x \in S$.
- Additive Inverse: The function defined point-wise by $(-f)(x) = -(f(x))$.
- Distributive Properties: $a(f + g)(x) = a(f(x) + g(x)) = af(x) + ag(x) = (af + ag)(x)$.

□

Question 8. Suppose $\mathcal{V}$ is a real vector space.

- The *complexification* of $\mathcal{V}$, denoted by $\mathcal{V}_{\mathbb{C}}$, equals $\mathcal{V} \times \mathcal{V}$. An element of $\mathcal{V}_{\mathbb{C}}$ is an ordered pair (u, v) , where $u, v \in \mathcal{V}$, but we write this as $u + iv$.
- Addition on $\mathcal{V}_{\mathbb{C}}$ is defined by

$$(u_1 + iv_1) + (u_2 + iv_2) = (u_1 + u_2) + i(v_1 + v_2)$$

for all $u_1, v_1, u_2, v_2 \in \mathcal{V}$.

- Complex scalar multiplication on $\mathcal{V}_{\mathbb{C}}$ is defined by

$$(a + bi)(u + iv) = (au - bv) + i(av + bu)$$

for all $a, b \in \mathbb{R}$ and all $u, v \in \mathcal{V}$.

Prove that with the definitions of addition and scalar multiplication as above, $\mathcal{V}_{\mathbb{C}}$ is a complex vector space.

Proof. Because $\mathcal{V}$ is a real vector space, it is inherently closed under its own addition and scalar multiplication. Therefore, $(u_1 + u_2) \in \mathcal{V}$, $(v_1 + v_2) \in \mathcal{V}$, $(au - bv) \in \mathcal{V}$, and $(av + bu) \in \mathcal{V}$, guaranteeing the operations always produce valid elements of $\mathcal{V}_{\mathbb{C}}$.

The following properties are also satisfied:

- Additive Identity: $0 + i0$ (since $0 \in \mathcal{V}$).
- Additive Inverse: $-(u) + i(-v)$ (since $\mathcal{V}$ contains inverses for u and v).
- Multiplicative Identity: $1 + i0$ acts as the scalar 1, yielding $1u - 0v + i(1v + 0u) = u + iv$.
- Associativity, Commutativity, and Distributivity: These expand algebraically and hold true because the vectors in $\mathcal{V}$ and scalars in $\mathbb{R}$ already obey these rules.

□

1.6 Exercise 1C

Question 1. For each of the following subsets of $\mathbf{F}^3$, determine whether it is a subspace of $\mathbf{F}^3$.

- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 + 2x_2 + 3x_3 = 4\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1x_2x_3 = 0\}$
- $\{(x_1, x_2, x_3) \in \mathbb{F}^3 : x_1 = 5x_3\}$

Proof. We only need to check for closure under addition and scalar multiplication and existence of a zero element. By this logic, (a),(d) are subspaces while (b),(c) are not. □

Question 2. Verify all assertions about subspaces below

- (a) If $b \in \mathbb{F}$, then

$$\{(x_1, x_2, x_3, x_4) \in \mathbb{F}^4 : x_3 = 5x_4 + b\}$$

is a subspace of $\mathbb{F}^4$ if and only if $b = 0$.

- (b) The set of continuous real-valued functions on the interval $[0, 1]$ is a subspace of $\mathbb{R}^{[0,1]}$.
 (c) The set of differentiable real-valued functions on $\mathbb{R}$ is a subspace of $\mathbb{R}^{\mathbb{R}}$.
 (d) The set of differentiable real-valued functions f on the interval $(0, 3)$ such that $f'(2) = b$ is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.
 (e) The set of all sequences of complex numbers with limit 0 is a subspace of $\mathbb{C}^{\infty}$.

Proof. (a) Forward direction (If it is a subspace, then $b = 0$): Every subspace must contain the additive identity, which in $\mathbb{F}^4$ is the zero vector $(0, 0, 0, 0)$. Substituting these coordinates into the defining equation yields $0 = 5(0) + b$, which simplifies strictly to $b = 0$.

Reverse direction (If $b = 0$, then it is a subspace): Setting $b = 0$ changes the subset condition to $x_3 = 5x_4$. Clearly, this is closed under addition, scalar multiplication and the zero vector $(0, 0, 0, 0)$ exists.

- (b) To prove that W is a subspace of $\mathbb{R}^{[0,1]}$, we must verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in $\mathbb{R}^{[0,1]}$ is the zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in [0, 1]$. Because constant functions are continuous on $\mathbb{R}$, the zero function is continuous on $[0, 1]$. Therefore, $\mathbf{0} \in W$, proving that W is non-empty.
 (b) Closure under vector addition: Let $f, g \in W$. By definition, f and g are both continuous functions on $[0, 1]$. From the properties of continuous functions in real analysis, the sum of two continuous functions is continuous. Therefore, the function $(f + g)(x) = f(x) + g(x)$ is continuous on $[0, 1]$, which implies $f + g \in W$.
 (c) Closure under scalar multiplication: Let $f \in W$ and $c \in \mathbb{R}$. Since f is a continuous function on $[0, 1]$, its multiplication by a scalar constant c results in a continuous function. Thus, the function $(cf)(x) = cf(x)$ is continuous on $[0, 1]$, meaning $cf \in W$.

Since W contains the zero vector and is closed under both addition and scalar multiplication, W is a subspace of $\mathbb{R}^{[0,1]}$.

- (c) To prove that W is a subspace of $\mathbb{R}^{\mathbb{R}}$, we verify the three conditions of the Subspace Criterion:

- (a) Presence of the zero vector: The zero vector in $\mathbb{R}^{\mathbb{R}}$ is the zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in \mathbb{R}$. Because constant functions are differentiable everywhere on $\mathbb{R}$ (with a derivative of 0), the zero function is differentiable. Therefore, $\mathbf{0} \in W$, proving that W is non-empty.
 (b) Closure under vector addition: Let $f, g \in W$. By definition, f and g are both differentiable functions on $\mathbb{R}$. From standard calculus theorems, the sum of two differentiable functions is differentiable, and its derivative is given by $(f + g)'(x) = f'(x) + g'(x)$. Therefore, the function $(f + g)$ is differentiable on $\mathbb{R}$, which implies $f + g \in W$.
 (c) Closure under scalar multiplication: Let $f \in W$ and $c \in \mathbb{R}$. Since f is a differentiable function on $\mathbb{R}$, its multiplication by a scalar constant c results in a differentiable function, with the derivative $(cf)'(x) = cf'(x)$. Thus, the function cf is differentiable on $\mathbb{R}$, meaning $cf \in W$.

Since W contains the zero vector and is closed under both vector addition and scalar multiplication, W is a subspace of $\mathbb{R}^{\mathbb{R}}$.

(d) Let $\mathcal{V} = \mathbb{R}^{(0,3)}$ and let $W_b = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = b\}$. We must prove that W_b is a subspace of $\mathcal{V}$ if and only if $b = 0$.

($\Rightarrow$) Assume W_b is a subspace of $\mathcal{V}$. Then it must contain the zero vector of $\mathcal{V}$, which is the zero function $\mathbf{0}$ defined by $\mathbf{0}(x) = 0$ for all $x \in (0,3)$. The derivative of the zero function is $\mathbf{0}'(x) = 0$ for all $x \in (0,3)$. Since $\mathbf{0} \in W_b$, it must satisfy the condition defining the set W_b , which means $\mathbf{0}'(2) = b$. Thus, $0 = b$.

($\Leftarrow$) Assume $b = 0$. We must show that $W_0 = \{f \in \mathcal{V} \mid f \text{ is differentiable on } (0,3) \text{ and } f'(2) = 0\}$ is a subspace. We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero function $\mathbf{0}$ is differentiable, and as shown above, $\mathbf{0}'(2) = 0$. Thus, $\mathbf{0} \in W_0$.
2. Closure under vector addition: Let $f, g \in W_0$. Both are differentiable on $(0,3)$, so their sum $f + g$ is also differentiable. By the linearity of the derivative, $(f + g)'(2) = f'(2) + g'(2) = 0 + 0 = 0$. Therefore, $f + g \in W_0$.
3. Closure under scalar multiplication: Let $f \in W_0$ and $c \in \mathbb{R}$. The function cf is differentiable, and its derivative at $x = 2$ is $(cf)'(2) = cf'(2) = c(0) = 0$. Therefore, $cf \in W_0$.

Since W_0 contains the zero vector and is closed under both addition and scalar multiplication, it is a subspace of $\mathcal{V}$.

Therefore, W_b is a subspace of $\mathbb{R}^{(0,3)}$ if and only if $b = 0$.

My thoughts

I don't know anything about complex sequences, but it is easy enough to wager a guess after a google search.

(e)

Let $\mathbb{C}^\infty$ be the vector space of all sequences of complex numbers, and let W be the set of all sequences of complex numbers with limit 0 [cite: 3]. We will prove that W is a subspace of $\mathbb{C}^\infty$.

We verify the three conditions of the subspace criterion:

1. Presence of the zero vector: The zero vector in $\mathbb{C}^\infty$ is the zero sequence, denoted by $\mathbf{0} = (0, 0, 0, \dots)$, where $z_n = 0$ for all $n \geq 1$. The limit of this constant sequence is $\lim_{n \rightarrow \infty} 0 = 0$. Therefore, $\mathbf{0} \in W$, meaning the subset is non-empty.
2. Closure under vector addition: Let $x = (x_n)$ and $y = (y_n)$ be sequences in W . By definition, $\lim_{n \rightarrow \infty} x_n = 0$ and $\lim_{n \rightarrow \infty} y_n = 0$. According to the algebraic limit theorems for sequences, the limit of their sum is given by $\lim_{n \rightarrow \infty} (x_n + y_n) = \lim_{n \rightarrow \infty} x_n + \lim_{n \rightarrow \infty} y_n = 0 + 0 = 0$. Thus, the sequence $x + y \in W$.
3. Closure under scalar multiplication: Let $x = (x_n) \in W$ and let $\lambda \in \mathbb{C}$. By definition, $\lim_{n \rightarrow \infty} x_n = 0$. Using the limit laws for scalar multiples, $\lim_{n \rightarrow \infty} (\lambda x_n) = \lambda \lim_{n \rightarrow \infty} x_n = \lambda \cdot 0 = 0$. Therefore, the sequence $\lambda x \in W$.

So W is a subspace of $\mathbb{C}^\infty$. □

Question 3. Show that the set of differentiable real-valued functions f on the interval $(-4, 4)$ such that $f'(-1) = 3f(2)$ is a subspace of $\mathbf{R}^{(-4,4)}$.

Proof. Let $W = \{f \in V \mid f \text{ is differentiable on } (-4, 4) \text{ and } f'(-1) = 3f(2)\}$. We verify that W satisfies the three conditions of a subspace:

1. **Additive identity:** The zero function, denoted by $\mathbf{0}$, where $\mathbf{0}(x) = 0$ for all $x \in (-4, 4)$, is differentiable. Its derivative is everywhere zero, so $\mathbf{0}'(-1) = 0$. Evaluating the other side gives $3\mathbf{0}(2) = 3(0) = 0$. Thus, $\mathbf{0}'(-1) = 3\mathbf{0}(2)$, meaning $\mathbf{0} \in W$.
2. **Closed under addition:** Let $f, g \in W$. Since f and g are differentiable, their sum $f + g$ is also differentiable. By the linearity of the derivative, we have:

$$(f + g)'(-1) = f'(-1) + g'(-1)$$

Because $f, g \in W$, we can substitute $f'(-1) = 3f(2)$ and $g'(-1) = 3g(2)$:

$$(f + g)'(-1) = 3f(2) + 3g(2) = 3(f(2) + g(2)) = 3(f + g)(2)$$

Therefore, $f + g \in W$.

3. **Closed under scalar multiplication:** Let $f \in W$ and $\lambda \in \mathbb{R}$. The scalar multiple λf is differentiable, and by the linearity of the derivative:

$$(\lambda f)'(-1) = \lambda f'(-1)$$

Since $f \in W$, we know $f'(-1) = 3f(2)$, yielding:

$$(\lambda f)'(-1) = \lambda(3f(2)) = 3(\lambda f(2)) = 3(\lambda f)(2)$$

Therefore, $\lambda f \in W$.

Since W contains the zero vector and is closed under both addition and scalar multiplication, W is a subspace of $\mathbb{R}^{(-4,4)}$. $\square$

Question 4. Suppose $b \in \mathbb{R}$. Show that the set of continuous real-valued functions f on the interval $[0, 1]$ such that $\int_0^1 f = b$ is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$.

Proof. Let $W_b = \left\{ f \in V \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = b \right\}$. We want to show that W_b is a subspace of V if and only if $b = 0$.

($\Rightarrow$) Assume W_b is a subspace of $\mathbb{R}^{[0,1]}$. Then W_b must contain the additive identity of $\mathbb{R}^{[0,1]}$, which is the zero function $\mathbf{0}$ defined by $\mathbf{0}(x) = 0$ for all $x \in [0, 1]$. The zero function is continuous, and its integral is:

$$\int_0^1 \mathbf{0} = \int_0^1 0 \, dx = 0$$

Since $\mathbf{0} \in W_b$, it must satisfy the condition defining the set, meaning $\int_0^1 \mathbf{0} = b$. Therefore, $b = 0$.

($\Leftarrow$) Assume $b = 0$. We must show that $W_0 = \left\{ f \in \mathbb{R}^{[0,1]} \mid f \text{ is continuous on } [0, 1] \text{ and } \int_0^1 f = 0 \right\}$ is a subspace. We verify the three conditions:

1. **Additive identity:** As shown above, the zero function $\mathbf{0}$ is continuous and $\int_0^1 \mathbf{0} = 0$. Thus, $\mathbf{0} \in W_0$.
2. **Closed under addition:** Let $f, g \in W_0$. Since both are continuous, their sum $f + g$ is also continuous. By the linearity of the definite integral:

$$\int_0^1 (f + g) = \int_0^1 f + \int_0^1 g = 0 + 0 = 0$$

Therefore, $f + g \in W_0$.

3. **Closed under scalar multiplication:** Let $f \in W_0$ and $\lambda \in \mathbb{R}$. The scalar multiple λf is continuous, and by the linearity of the integral:

$$\int_0^1 (\lambda f) = \lambda \int_0^1 f = \lambda(0) = 0$$

Therefore, $\lambda f \in W_0$.

Thus, W_b is a subspace of $\mathbb{R}^{[0,1]}$ if and only if $b = 0$. □

Question 5. Is $\mathbb{R}^2$ a subspace of the complex vector space $\mathbb{C}^2$?

Proof. No, $\mathbb{R}^2$ is not a subspace of the complex vector space $\mathbb{C}^2$.

To be a subspace of the complex vector space $\mathbb{C}^2$, the subset $\mathbb{R}^2$ must be closed under scalar multiplication for all scalars in the field $\mathbb{C}$.

Consider the vector $v = (1, 0) \in \mathbb{R}^2$ and the complex scalar $\lambda = i \in \mathbb{C}$. Multiplying the vector by the scalar yields:

$$\lambda v = i(1, 0) = (i, 0)$$

Since $i \notin \mathbb{R}$, the resulting vector $(i, 0)$ is not in $\mathbb{R}^2$. □

Question 6. (a) Is $\{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$ a subspace of $\mathbb{R}^3$?
 (b) Is $\{(a, b, c) \in \mathbb{C}^3 : a^3 = b^3\}$ a subspace of $\mathbb{C}^3$?

Proof. **(a) Yes, it is a subspace.** Let $W = \{(a, b, c) \in \mathbb{R}^3 : a^3 = b^3\}$. For real numbers, the function $f(x) = x^3$ is injective, so $a^3 = b^3$ implies $a = b$. Thus, W can be rewritten as $W = \{(a, a, c) \mid a, c \in \mathbb{R}\}$. We verify the subspace conditions:

1. Additive identity: The zero vector $(0, 0, 0)$ is in W since $0 = 0$.
2. Closed under addition: Let $v_1 = (a_1, a_1, c_1)$ and $v_2 = (a_2, a_2, c_2)$ be in W . Their sum $v_1 + v_2 = (a_1 + a_2, a_1 + a_2, c_1 + c_2)$ is in W because its first two coordinates are equal.
3. Closed under scalar multiplication: Let $v = (a, a, c) \in W$ and $\lambda \in \mathbb{R}$. The scalar multiple $\lambda v = (\lambda a, \lambda a, \lambda c)$ is in W because its first two coordinates are equal.

Therefore, W is a subspace of $\mathbb{R}^3$.

(b) No, it is not a subspace. Consider the vectors $v_1 = (1, 1, 0)$ and $v_2 = (\omega, 1, 0)$. Both vectors are in U because $1^3 = 1^3$ and $\omega^3 = 1 = 1^3$. Their sum is:

$$v_1 + v_2 = (1 + \omega, 2, 0)$$

For this sum to be in U , its coordinates must satisfy $a^3 = b^3$, meaning we must have $(1 + \omega)^3 = 2^3 = 8$. Expanding the left side:

$$(1 + \omega)^3 = 1^3 + 3(1^2)\omega + 3(1)\omega^2 + \omega^3 = 1 + 3\omega + 3\omega^2 + 1 = 2 + 3(\omega + \omega^2)$$

Since $1 + \omega + \omega^2 = 0$ for cube roots of unity, we know $\omega + \omega^2 = -1$. Thus:

$$(1 + \omega)^3 = 2 + 3(-1) = -1$$

Since $-1 \neq 8$, the vector $v_1 + v_2 \notin U$. Because U is not closed under addition, it is not a subspace of $\mathbb{C}^3$. □

Question 7. Prove or give a counterexample: If U is a nonempty subset of $\mathbb{R}^2$ such that U is closed under addition and under taking additive inverses (meaning $-u \in U$ whenever $u \in U$), then U is a subspace of $\mathbb{R}^2$.

Proof. The statement is false. We provide a counterexample.

Let $U = \mathbf{Z}^2 = \{(m, n) \mid m, n \in \mathbf{Z}\}$, which is the set of all vectors in $\mathbf{R}^2$ with integer coordinates.

We verify that U satisfies the given conditions:

1. **U is nonempty:** The vector $(0, 0) \in U$ because 0 is an integer.
2. **Closed under addition:** Let $u = (m_1, n_1) \in U$ and $v = (m_2, n_2) \in U$. Their sum is $u + v = (m_1 + m_2, n_1 + n_2)$. Since the sum of two integers is an integer, $u + v \in U$.
3. **Closed under additive inverses:** Let $u = (m, n) \in U$. Its additive inverse is $-u = (-m, -n)$. Since the negative of an integer is an integer, $-u \in U$.

However, U is not a subspace of $\mathbf{R}^2$ because it is not closed under scalar multiplication over the field $\mathbf{R}$.

For a specific counterexample, let $u = (1, 1) \in U$ and let the scalar $\lambda = \frac{1}{2} \in \mathbf{R}$. Multiplying them yields:

$$\lambda u = \frac{1}{2}(1, 1) = \left(\frac{1}{2}, \frac{1}{2}\right)$$

Because $\frac{1}{2}$ is not an integer, the resulting vector $\left(\frac{1}{2}, \frac{1}{2}\right) \notin U$.

Since U is not closed under scalar multiplication, it fails the subspace criterion. Therefore, the statement is false. □